1. WHAT IS ORPO?

- **ORPO** = Odds Ratio Preference Optimization; it is a preference-optimization method for fine-tuning language models.
- ORPO combines the supervised fine-tuning (SFT) objective with an odds-ratio preference objective in a single training stage.
- It learns from preferred and rejected responses without requiring a separately train model.

2. WHAT IS A PREFERENCE PAIR?

A preference pair contains a prompt and two responses to that prompt: one preferred and one rejected. These pairs are prepared as training data before ORPO fine-tuning.

✓ **Preferred answer**
(higher quality / better aligned)

✗ **Rejected answer**
(lower quality / less suitable)

End-to-End Pipeline: From Preference Pairs to Runtime Quality Gating

1 Task Prompt

Explain photosynthesis in simple words.

User instruction
The user gives a q tion or task, with any needed context.

Task prompt

2 Generate Two Responses

✓ **Preferred:**
Plants use sunlight to make food from water and carbon dioxide.

✗ **Rejected:**
Photosynthesis is the process by which atoms form nuclear explosions.

Preference pair

3 Single-Stage ORPO

ORPO fine-tuning
Learns from (y_w, y_l) using combined SFT and odds-ratio preference losses.

The output is a fine-tuned model. No separate reward model is required.

Trained model

4 Quality Gate

Checks the generated response using separate runtime rules.

Example checks:
- Task relevance
- Coherence
- Applicable safety requirements

Pass / Fail

5 Final Output

✓ **Approved**
Passes the defined checks.

✗ **Rejected or revised**
Action follows the defined rules.

ORPO vs. Traditional PPO-Based RLHF

ORPO

- ✓ Combines SFT and preference optimization in one stage.
- ✓ No separate reward model.
- ✓ Fewer separate training components.

Traditional PPO-Based RLHF

- ✓ Typically uses SFT, a reward model, and PPO.
- ✓ Requires a separate reward model.
- ✓ More involved multi-stage pipeline.

Why Choose ORPO for This Project?

- **Direct preference learning:** learns from preferred and rejected responses without a separate reward-model training step.
- **Simpler training workflow:** fewer separate training components make the workflow easier to implement and explain.
- **Runtime validation:** the separate quality gate can check relevance, coherence, and applicable safety requirements. ORPO itself does not guarantee that every response is safe or correct.